ASDS 5301, Project 4: Statistical Theory and Application

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Introduction:
This study aims to find examine factors such as the following:

- Study_Hours: Average number of hours spent studying per week
- Class_Attendance: Percentage of classes attended
- Final_Score: Final exam score

For 200 students to see how these factors affect their final exam scores in their statistics course. For this study, we will be using a significance level of 0.01.

Loading and Prepping the data:

```
/* Generated Code (IMPORT) //Source File: student (1).xlsx //Source Path: /home/u63988430
//Code generated on: 05/11/2026, 03:46 */

%web_drop_table(WORK.IMPORT);

FILENAME REFFILE '/home/u63988430/student (1).xlsx';

PROC IMPORT DATAFILE=REFFILE DBMS=XLSX OUT=WORK.IMPORT REPLACE;
GETNAMES=YES; RUN;

PROC CONTENTS DATA=WORK.IMPORT; RUN;

%web_open_table(WORK.IMPORT);
```

The aforementioned code created a table that we will be using for testing our hypotheses.

```
DATA WORK.student_analysis;
    SET WORK.IMPORT;
```

```
    /* Part D variable */
    IF Final_Score >= 85 THEN Pass = 1;
    ELSE IF Final_Score < 85 AND NOT MISSING(Final_Score) THEN Pass = 0;

    /* Bonus variable */
    IF Class_Attendance >= 85 THEN Attendance_Group = 'High (>=85%)';
    ELSE IF Class_Attendance < 85 AND NOT MISSING(Class_Attendance) THEN
Attendance_Group = 'Low (<85%)';
RUN;
```

This code creates the class for the bonus question.

PART A: Exploratory Data Analysis

1. Summary Statistics:

```
/* A1: Summary Statistics for all continuous variables */
TITLE "Part A1: Descriptive Summary Statistics";
PROC UNIVARIATE DATA=WORK.student_analysis;
    VAR Final_Score Study_Hours Class_Attendance;
RUN;
TITLE;
```

Part A1: Descriptive Summary Statistics

The UNIVARIATE Procedure
Variable: Final_Score (Final_Score)

Moments			
N	200	Sum Weights	200
Mean	86.90215	Sum Observations	17380.43
Std Deviation	9.72079855	Variance	94.4939245
Skewness	-0.6017048	Kurtosis	0.01766682
Uncorrected SS	1529201.03	Corrected SS	18804.291
Coeff Variation	11.1859126	Std Error Mean	0.68736426

Part A1: Descriptive Summary Statistics

The UNIVARIATE Procedure
Variable: Study_Hours (Study_Hours)

Moments			
N	200	Sum Weights	200
Mean	11.99705	Sum Observations	2399.41
Std Deviation	3.01275826	Variance	9.07671236
Skewness	0.09346105	Kurtosis	-0.0883829
Uncorrected SS	30592.1075	Corrected SS	1806.26576
Coeff Variation	25.1124924	Std Error Mean	0.21303418

Part A1: Descriptive Summary Statistics

The UNIVARIATE Procedure
Variable: Class_Attendance (Class_Attendance)

Moments			
N	200	Sum Weights	200
Mean	84.35985	Sum Observations	16871.97
Std Deviation	9.60615645	Variance	92.2782417
Skewness	-0.2209707	Kurtosis	-0.3296441
Uncorrected SS	1441680.23	Corrected SS	18363.3701
Coeff Variation	11.3871189	Std Error Mean	0.67925784

2. Summary statistics (mean, variance, correlation) condense distributions into single scalar values and can completely mask non-linear patterns, clustering, heteroscedasticity, and influential leverage points. A scatterplot confirms whether the relationship is actually linear

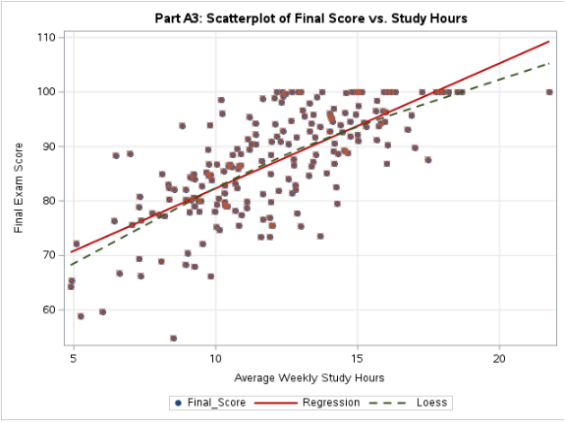
before fitting a best fit line, and flags single extreme outliers that could disproportionately torque regression slopes.

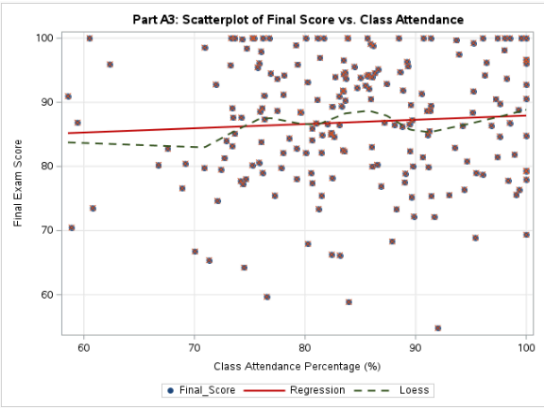
3. Scatterplots:

```
ODS GRAPHICS ON;

TITLE "Part A3: Scatterplot of Final Score vs. Study Hours";
PROC SGPLOT DATA=WORK.student_analysis;
   SCATTER X=Study_Hours Y=Final_Score /
      MARKERATTRS=(COLOR=cx1F497D SYMBOL=circlefilled SIZE=7);
   REG X=Study_Hours Y=Final_Score /
      LINEATTRS=(COLOR=cxC00000 THICKNESS=2);
   LOESS X=Study_Hours Y=Final_Score /
      LINEATTRS=(COLOR=cx385723 THICKNESS=2 PATTERN=dash);
   XAXIS LABEL="Average Weekly Study Hours" GRID;
   YAXIS LABEL="Final Exam Score" GRID;
RUN;

TITLE "Part A3: Scatterplot of Final Score vs. Class Attendance";
PROC SGPLOT DATA=WORK.student_analysis;
   SCATTER X=Class_Attendance Y=Final_Score /
      MARKERATTRS=(COLOR=cx1F497D SYMBOL=circlefilled SIZE=7);
   REG X=Class_Attendance Y=Final_Score /
      LINEATTRS=(COLOR=cxC00000 THICKNESS=2);
   LOESS X=Class_Attendance Y=Final_Score /
      LINEATTRS=(COLOR=cx385723 THICKNESS=2 PATTERN=dash);
   XAXIS LABEL="Class Attendance Percentage (%)" GRID;
   YAXIS LABEL="Final Exam Score" GRID;
RUN;
TITLE;
```





4. Conclusion:
Based on the graphs, Study Hours is a much more significant predictor for Final Scores than Class Attendance. This is because the slope for Study Hours is much steeper than the one for Class Attendance and the points in Study Hours are much closer to the line as well, indicating a stronger relationship.

Part B: Multiple Linear Regression

1. Multiple Linear Regression Model:

```
TITLE "Part B: Full Multiple Linear Regression Model & Diagnostics";
PROC REG DATA=WORK.student_analysis
   PLOTS(MAXPOINTS=NONE)=(DIAGNOSTICS RESIDUALS(SMOOTH)
OBSERVEDBYPREDICTED);
   MODEL Final_Score = Study_Hours Class_Attendance / ALPHA=0.01 CLB VIF;
RUN;
QUIT;
TITLE;
```

Part B: Full Multiple Linear Regression Model & Diagnostics

The REG Procedure
Model: MODEL1
Dependent Variable: Final_Score Final_Score

Number of Observations Read	200
Number of Observations Used	200

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	9763.32560	4881.66280	106.37	<.0001
Error	197	9040.96537	45.89323		
Corrected Total	199	18804			

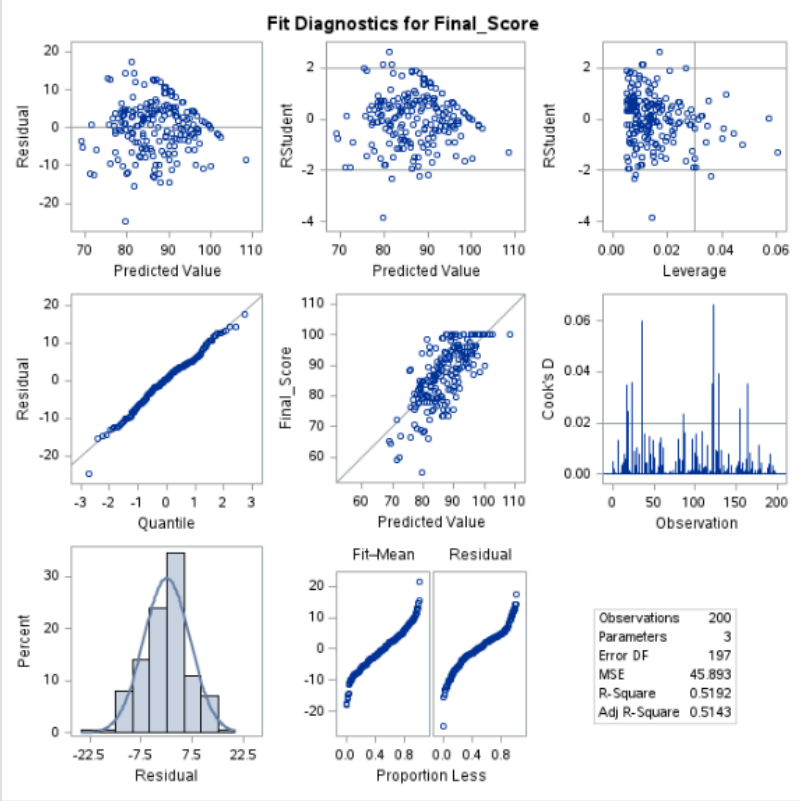
Root MSE	6.77445	R-Square	0.5192
Dependent Mean	86.90215	Adj R-Sq	0.5143
Coeff Var	7.79550		

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation	99% Confidence Limits	
Intercept	Intercept	1	49.43630	4.77812	10.35	<.0001	0	37.00834	61.86426
Study_Hours	Study_Hours	1	2.32032	0.15974	14.53	<.0001	1.00434	1.90482	2.73581
Class_Attendance	Class_Attendance	1	0.11414	0.05010	2.28	0.0238	1.00434	-0.01617	0.24445

Final Score = 49.436 + 2.320 (Study Hours) + 0.114 (Class_Attendance)

2. Diagnostic Plots:



The diagnostic plots confirm that the multiple linear regression model is appropriate, as the residuals display a random, uniform spread around zero without noticeable curvature or heteroscedasticity.

3. Estimated coefficient for Study_Hours:

Holding class attendance constant, each additional hour studied per week is associated with an estimated average increase of 2.32 points on the final exam.

a. Hypotheses: H0: b1 = 0 vs. Ha: b1 != 0

b. Test Statistic, P-value, and Conclusion (alpha = 0.01):

Test statistic: t = 14.53, p-value < 0.0001.

Conclusion: Since p < 0.0001 < 0.01, we reject H0; weekly study hours are a statistically significant positive predictor of final exam score.

4. Estimated coefficient for Class_Attendance:

Holding weekly study hours constant, each 1% increase in class attendance is associated with an estimated average increase of 0.11 points on the final exam.

a. Hypotheses: H0: b2 = 0 vs. Ha: b2 != 0

b. Test Statistic, P-value, and Conclusion (alpha = 0.01):

Test statistic: t = 2.28, p-value = 0.0238.

Conclusion: Since p = 0.0238 >= 0.01, we fail to reject H0; class attendance is not a statistically significant predictor of final exam score at alpha = 0.01 after controlling for study hours.

5. Predictor Significance and Model Removal

Class_Attendance is statistically insignificant at alpha = 0.01 since its p-value (0.0238) exceeds 0.01.

Removing Class_Attendance is reasonable to maintain model parsimony and avoid overfitting, as doing so simplifies the model without meaningfully sacrificing explanatory power.

Part C: Model Reduction

1. Compare R-Square for Full and Reduced Models:

```
/*Class_Attendance is insignificant */

TITLE "Part C: Reduced Model (Study_Hours Only)";

PROC REG DATA=WORK.student_analysis;

    MODEL Final_Score = Study_Hours / ALPHA=0.01;

RUN;

QUIT;

TITLE;
```

Part C: Reduced Model (Study_Hours Only)

The REG Procedure
Model: MODEL1
Dependent Variable: Final_Score Final_Score

Number of Observations Read	200
Number of Observations Used	200

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	9525.12055	9525.12055	203.25	<.0001
Error	198	9279.17042	46.86450		
Corrected Total	199	18804			

Root MSE	6.84576	R-Square	0.5065
Dependent Mean	86.90215	Adj R-Sq	0.5040
Coeff Var	7.87756		

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	59.35232	1.99214	29.79	<.0001
Study_Hours	Study_Hours	1	2.29638	0.16108	14.26	<.0001

The full model has an R-Square of 0.5192 (Adjusted R-Square = 0.5143), while the reduced model has an R-Square of 0.5065 (Adjusted R-Square = 0.5040). Removing Class_Attendance results in a negligible decrease in explained variance of only 0.0127 (about 1.27%). This demonstrates that Study_Hours accounts for almost all of the explanatory capability in predicting Final_Score.

2: Does Removing the Insignificant Variable Meaningfully Harm the Model?

No, removing Class_Attendance does not meaningfully harm the model because the drop in R-Square is minimal and the Root MSE only slightly increases from 6.77 to 6.85. The residual error variance remains virtually identical (MSE moves from 45.89 to 46.86), showing no practical loss in predictive accuracy. Consequently, the model retains its predictive strength without relying on an insignificant predictor at alpha = 0.01.

3: Recommended Model and Why

I recommend the reduced model featuring Study_Hours as the sole predictor. It adheres to the principle of parsimony by removing an insignificant variable at alpha = 0.01 while capturing over 50.6% of the variation in final exam scores. A simpler model avoids unnecessary parameter estimation, prevents potential overfitting, and provides an easier, more actionable framework for academic intervention.

Part D: Proportion Inference

1. Creating Binary Pass Variable:

```
/* At alpha = 0.01 */

TITLE "Part D: Proportion Inference Test (p0 = 0.65, Alpha = 0.01)";

PROC FREQ DATA=WORK.student_analysis;

    TABLES Pass / BINOMIAL(P=0.65 LEVEL='1') ALPHA=0.01;

RUN;


/* At alpha = 0.05 */

TITLE "Part D: Proportion Inference Test (p0 = 0.65, Alpha = 0.05)";

PROC FREQ DATA=WORK.student_analysis;

    TABLES Pass / BINOMIAL(P=0.65 LEVEL='1') ALPHA=0.05;

RUN;

TITLE;
```

Part D: Proportion Inference Test (p0 = 0.65, Alpha = 0.01)

The FREQ Procedure

Pass	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	82	41.00	82	41.00
1	118	59.00	200	100.00

Binomial Proportion	
Pass = 1	
Proportion	0.5900
ASE	0.0348
99% Lower Conf Limit	0.5004
99% Upper Conf Limit	0.6796
Exact Conf Limits	
99% Lower Conf Limit	0.4966
99% Upper Conf Limit	0.6790

Test of H0: Proportion = 0.65	
ASE under H0	0.0337
Z	-1.7790
One-sided Pr < Z	0.0376
Two-sided Pr > |Z|	0.0752

Sample Size = 200

Part D: Proportion Inference Test (p0 = 0.65, Alpha = 0.05)

The FREQ Procedure

Pass	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	82	41.00	82	41.00
1	118	59.00	200	100.00

Binomial Proportion	
Pass = 1	
Proportion	0.5900
ASE	0.0348
95% Lower Conf Limit	0.5218
95% Upper Conf Limit	0.6582
Exact Conf Limits	
95% Lower Conf Limit	0.5184
95% Upper Conf Limit	0.6589

Test of H0: Proportion = 0.65	
ASE under H0	0.0337
Z	-1.7790
One-sided Pr < Z	0.0376
Two-sided Pr > |Z|	0.0752

Sample Size = 200

Hypotheses

H0: p >= 0.65 (The true pass rate is at least 65%)

Ha: p < 0.65 (The true pass rate is less than 65%).

a) Test Statistic and Meaning

The test statistic is Z = -1.7790. It measures how many standard errors the observed sample pass proportion (p = 0.59) falls below the hypothesized proportion of 0.65 under the null hypothesis.

b) Hypothesis Test at alpha = 0.01

At alpha = 0.01, the one-sided p-value is 0.0376 (Pr < Z). Since 0.0376 >= 0.01, we fail to reject the null hypothesis H0. There is insufficient evidence at the 1% significance level to dispute the university's claim that at least 65% of students pass the course.

c) Hypothesis Test at alpha = 0.05

At alpha = 0.05, the one-sided p-value remains 0.0376 (Pr < Z). Since 0.0376 < 0.05, we reject the null hypothesis H0. There is statistically significant evidence at the 5% significance level to conclude that the true passing proportion is less than 65%, contradicting the university's claim.

Bonus Question:

```
TITLE "Bonus Part 3: Pass Rates by Attendance Group";

PROC FREQ DATA=WORK.student_analysis;

    TABLES Attendance_Group * Pass / CHISQ RELRISK NOROW NOCOL;

RUN;

TITLE;
```

Bonus 1: Random Sampling vs. Random Assignment

Random sampling ensures sample statistics are unbiased estimates of population parameters by eliminating selection bias. Random assignment balances lurking confounders across groups to allow valid causal inference rather than simple association.

Bonus 2: Influence of Sample Size

A larger sample size reduces standard error, narrowing confidence intervals and improving the accuracy of the normal approximation under the Central Limit Theorem. It also increases statistical power, lowering the risk of a Type II error when testing hypotheses.

Bonus 3.a: Frequency Table

Attendance_Group by Pass: High (>=85%): Fail (0) = 36 (18.00%), Pass (1) = 57 (28.50%), Total = 93 (46.50%) Low (<85%): Fail (0) = 46 (23.00%), Pass (1) = 61 (30.50%), Total = 107 (53.50%) Overall Total: Fail (0) = 82 (41.00%), Pass (1) = 118 (59.00%), Total = 200 (100.00%)

Bonus 3.b: Pass Proportions in Each Group

High Attendance Pass Rate: 57 / 93 = 0.6129 (61.29%). Low Attendance Pass Rate: 61 / 107 = 0.5701 (57.01%).

Bonus 3.c: Relation to Earlier Regression Findings

The two pass rates are nearly identical, and the Chi-Square test confirms no statistically significant association between attendance and passing (Chi-Square = 0.3770, p = 0.5392). This directly mirrors the regression findings, where Class_Attendance was statistically insignificant at alpha = 0.01 and contributed negligible explanatory power.

AI acknowledgement: Gemini was used to plot the graphs and tables.